Thermal Expansion

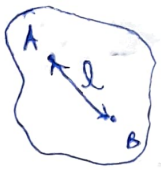
→ Solids and liquids expands slightly on increasing temperature.

Temp↑ ⇒ Thermal agitation of molecules ↑ ⇒ intermolecular separation ↑

⇒ length ↑, width ↑, height ↑, area ↑, volume ↑, density ↓

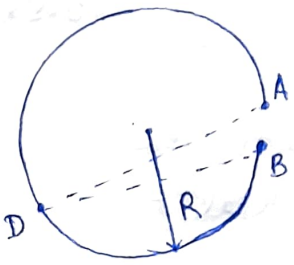
→ Thermal expansions are important to consider while constructing our structures such as flyovers, railway tracks etc.

* On increasing temperature, shape of object remains unchanged, only sizes has increased.
The object becomes photographic enlargement of itself.
* On increasing temperature, separation b/w any pair of points on an object will increase.



As T↑ ⇒ l↑

Q

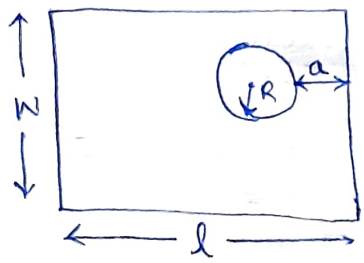


On ↑ temp which quantities will ↑?

(a) R ✓
(b) AD ✓
(c) BD ✓
(d) AB ✓

} As on ↑ temp, sep. b/w any two pts ↑.

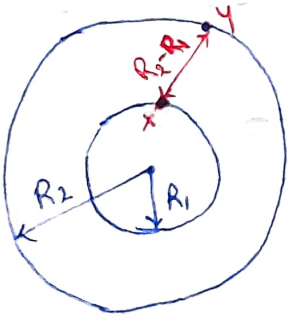
Q



On ↑ temp which quantities will ↑?

(a) l ✓
(b) w ✓
(c) R ✓
(d) a ✓

Q



On ↑ temp which quantities will ↑?

(1) R_1 ✓
(2) R_2 ✓
(3) $R_1 + R_2$ ✓
(4) $R_2 - R_1$ ✓ (As XY ↑)

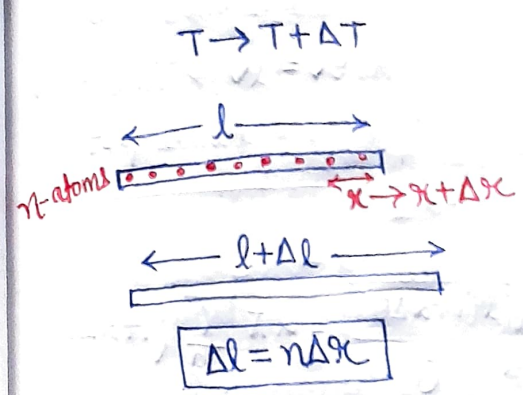
Note: $dR_2 > dR_1$

ie. R_2 ↑ more as compared to R_1 due to bigger dimensions initially

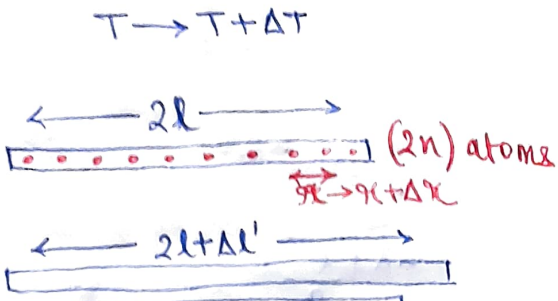
Linear Expansion:

→ Linear expansion is change in length due to change in temperature.

→ As $T \uparrow \Rightarrow l \uparrow$



$$\Delta l = n\Delta x$$



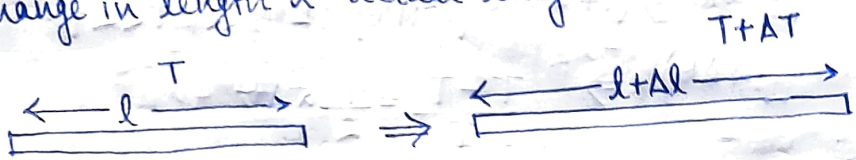
$$\Delta l' = (2n)\Delta x$$

$$\Rightarrow \Delta l' = 2(\Delta l)$$

∴ We conclude, $\boxed{\Delta l \propto l}$

→ change in length $\propto$ actual length

Now,



$\Delta l \propto l$

Also, $\Delta l \propto \Delta T$

∴ * $\boxed{\Delta l = \alpha l \Delta T}$

α = Coefficient of linear expansion

- → property of material
- → unit = $1/^\circ C$ or $1/\text{kelvin}$
- → Dimension = $[K^{-1}]$

$$l_{final} = l + \Delta l = l + \alpha l \Delta T = l(1 + \alpha \Delta T)$$

Fractional change in length = $\frac{\Delta l}{l} = \frac{\alpha l \Delta T}{l} = \alpha \Delta T$

% change in length = $\frac{\Delta l}{l} \times 100 = \frac{\alpha l \Delta T}{l} \times 100 = (\alpha \Delta T \times 100)$

Q Rod of $l = 3m$

$\alpha = 4 \times 10^{-4} /^\circ C$

Temp = $10^\circ C \rightarrow 30^\circ C$

Find ① Change in length

② Final length

③ % change in length

① $\Delta l = \alpha l \Delta T$

$= \frac{4 \times 10^{-4}}{^\circ C} \times 3m \times 20^\circ C$

$= 24 \times 10^{-3} m$

$= 24mm$

$= 2.4cm$

② $l_{final} = l + \Delta l$

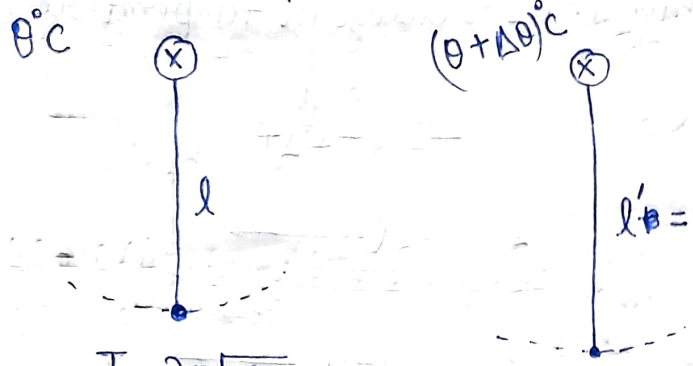
$= 3m + 0.024m$

$= 3.024m$

③ % change in $l = \frac{\Delta l}{l} \times 100$

$\frac{0.024m}{3m \times 1000} \times 100 = 0.8\%$

Change in Time period of a simple pendulum:



As θ↑ ⇒ l↑ ⇒ Time period↑
⇒ Clock becomes slow
As θ↓ ⇒ l↓ ⇒ Time period↓
⇒ Clock becomes fast

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T' = 2\pi\sqrt{\frac{l'}{g}} = 2\pi\sqrt{\frac{l(1+\alpha\Delta\theta)}{g}} = 2\pi\sqrt{\frac{l}{g}}(1+\alpha\Delta\theta)^{1/2}$$

$$\therefore \boxed{T' = T(1+\alpha\Delta\theta)^{1/2}}$$

Using binomial approx. $\Rightarrow (1+x)^n \approx (1+nx)$ if $x \ll 1$

$$T' = T\left(1+\frac{1}{2}\alpha\Delta\theta\right) \Rightarrow \frac{T'-T}{T} = \frac{1}{2}\alpha\Delta\theta$$

$\therefore$ Fractional change in time period $= \frac{\Delta T}{T} = \frac{1}{2}\alpha\Delta\theta$

Q A simple pendulum gives correct time at 20°C. Its wire is made up of iron whose $\alpha = 1.2\times10^{-6}/°C$. How much slow/fast this pendulum clock is, in 1 summer day where Temperature is 40°C.

Fractional change in Time period $= \frac{1}{2}\alpha\Delta\theta = \frac{1}{2}\times\frac{1.2\times10^{-6}}{°C}\times 20°C$

$$\therefore \frac{\Delta T}{T} = 1.2\times10^{-5}$$

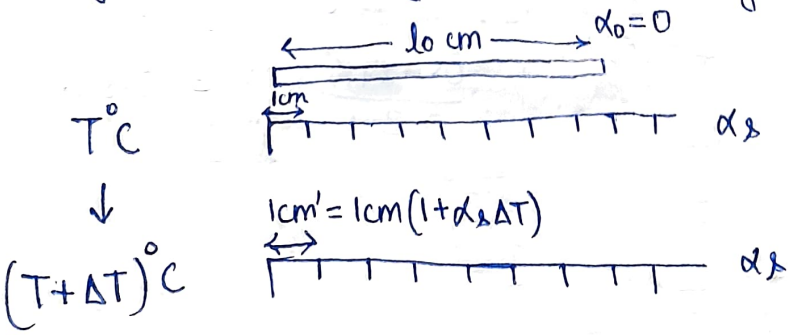
$\Rightarrow$ In 1 sec $\Rightarrow 1.2\times10^{-5}$ sec slow

In $24\times60\times60 \Rightarrow 1.2\times10^{-5}\times24\times60\times60$

$= 1036.8\times10^{-3}$ sec $= 1.036$ sec slow (in 24 hrs)

Measuring length using metallic scale:

① Object is not expanding, scale is expanding on increasing temperature:



$$\text{Reading} = \frac{l_o\ cm}{1\ cm} = l_o$$

$$\text{New Reading} = \frac{l_o\ cm}{1cm(1+\alpha_s\Delta T)} = \frac{l_o}{(1+\alpha_s\Delta T)} < l_o$$

$\therefore$ New reading is smaller.

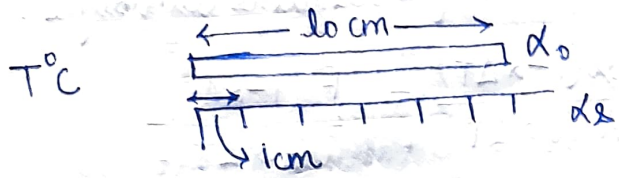
Q An object having $\alpha_o = 0$ have length 20cm reading if measured by a scale ($\alpha_s = 2\times10^{-5}/°C$). What will be reading if temp. increases by 20°C ?

$$\text{Reading} = \frac{l_o}{1+\alpha\Delta T} = \frac{20\,cm}{(1+2\times10^{-5}\times20)} = \frac{20\,cm}{(1+4\times10^{-4})} = 20\,cm\,(1+4\times10^{-4})^{-1}$$

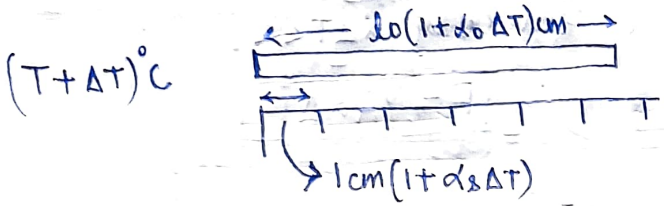
$$= 20\,cm\,(1-4\times10^{-4}) = 20\,cm\,(1-0.0004) = 20\,cm\,(0.9996) = 19.992\,cm$$

(using $(1+x)^n \approx (1+nx)$)

② If both object and scale are expanding :



$$\text{Reading} = \frac{l_o\,cm}{1\,cm} = l_o$$



$$\text{Reading} = \frac{l_o(1+\alpha_o\Delta T)}{1(1+\alpha_s\Delta T)}$$

$$\therefore \boxed{\text{Reading} = \frac{l_o(1+\alpha_o\Delta T)}{(1+\alpha_s\Delta T)}}$$

Let Temp↑ ⇒ ΔT = +ve

① If $\alpha_o = \alpha_s$ ⇒ Reading unchanged

② If $\alpha_o > \alpha_s$ ⇒ Reading increases

③ If $\alpha_o < \alpha_s$ ⇒ Reading decreases

Now,

$$\text{Reading} = l_o(1+\alpha_o\Delta T)(1+\alpha_s\Delta T)^{-1} = l_o(1+\alpha_o\Delta T)(1-\alpha_s\Delta T)$$

$$= l_o(1+\alpha_o\Delta T - \alpha_s\Delta T - \alpha_o\alpha_s\Delta T) \quad (\alpha_o\alpha_s\Delta T \to \text{very small}) \quad = l_o(1+(\alpha_o-\alpha_s)\Delta T)$$

$$\therefore \boxed{\text{Reading} = l_o(1+(\alpha_o-\alpha_s)\Delta T)}$$

Q 

$T = 20°C$ Reading = 40cm

what will be reading at 40°C ?

$$\text{Reading} = l_o(1+(\alpha_o-\alpha_s)\Delta T)$$

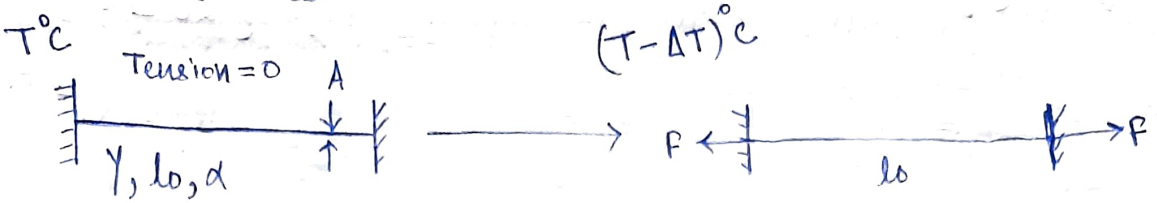
$$= 40\,cm\,(1+2\times10^{-4}\times20) = 40\,cm\,(1+4\times10^{-3})$$

$$= 40\,cm\,(1+0.004)$$

$$= 40\,cm\,(1.004)$$

$$= 40.16\,cm$$

Thermal Stress:



$\Delta l = l_o - l_o(1-\alpha\Delta T)$

$\Delta l = l_o \alpha \Delta T$

On decreasing temp. length of wire should have also decreased.

$l_{actual} = l_o(1-\alpha\Delta T)$

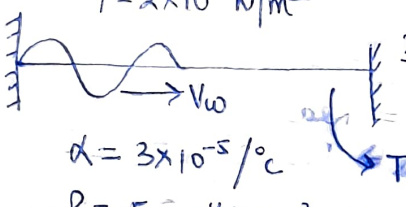
But the walls apply a force due to which length remains unchanged.

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta l/l} = \frac{Fl}{A\Delta l} = \frac{F\,l_o(1-\alpha\Delta T)}{A\cdot l_o\alpha\Delta T} = \frac{F(1-\alpha\Delta T)}{A\alpha\Delta T}$$

Tension in wire $\Rightarrow F = \frac{YA\alpha\Delta T}{(1-\alpha\Delta T)}$, Generally $\alpha\Delta T \ll 1$

$$\therefore \boxed{F = YA\alpha\Delta T}$$

Q $Y = 2\times10^{11}\ N/m^2$



$\alpha = 3\times10^{-5}/°C$

$\rho = 5\times10^{4}\ kg/m^3$

At 30°C there is no tension in wire. If temperature decreases to 20°C, what will be speed of wave on the wire?

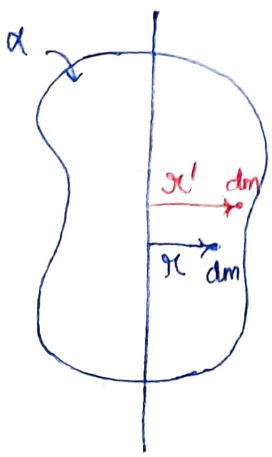
$$v_w = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{YA\alpha\Delta T}{\mu}} = \sqrt{\frac{Y\alpha\Delta T}{\rho}} = \sqrt{\frac{2\times10^{11}\times3\times10^{-5}\times10}{5\times10^{4}}}$$

$\left(\frac{\mu}{A} = \frac{m/l}{A} = \frac{m}{Al} = \frac{m}{V} = \rho\right)$

$$= \sqrt{\frac{20\times3}{5}\times10^{3}} = 20\sqrt{3}\ m/s$$

Effect of temperature on moment of Inertia:

→ Temp↑ ⇒ Mass will get farther from axis ⇒ MI↑



At $T°C \Rightarrow I_o = \int dm x^2$

At $(T+\Delta T)°C \Rightarrow I = \int dm x'^2 = \int dm x^2(1+\alpha\Delta T)^2$

$= (1+\alpha\Delta T)^2 \int dm x^2 = (1+\alpha\Delta T)^2 I_o$

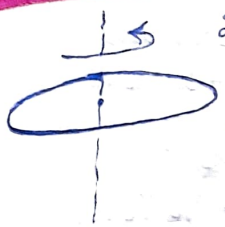
$$\therefore \boxed{I = (1+\alpha\Delta T)^2 I_o}$$

If $\alpha\Delta T \ll 1$

$$\Rightarrow \boxed{I = I_o(1+2\alpha\Delta T)}$$

$$\boxed{\Delta I = I - I_o = I_o(2\alpha\Delta T)}$$

8



$20^\circ C$

$\alpha = 5\times10^{-4}/^\circ C$

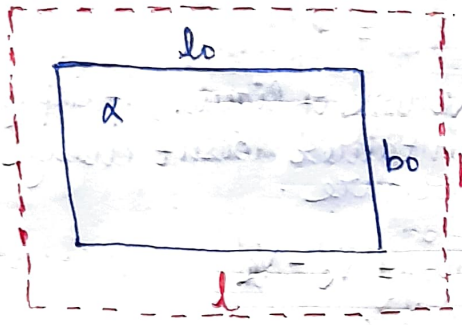
$I_0 = 2 kgm^2$

What will be moment of inertia at $0^\circ C$?

$$I = I_0(1+2\alpha\Delta T) = 2kgm^2\left(1+2\times\frac{5\times10^{-4}}{^\circ C}\times(-20^\circ C)\right) = 2kgm^2(1-0.02)$$

$$I = 2kgm^2 * 0.98 = 1.96 kgm^2$$

Areal Expansion / Superficial Expansion:



At $T_0^\circ C \Rightarrow A_0 = l_o b_o$

At $(T_0+\Delta T)^\circ C \Rightarrow A = lb$

$$A = l_o b_o(1+\alpha\Delta T)^2$$

$$A = A_0(1+\alpha\Delta T)^2$$

$$A = A_0(1+2\alpha\Delta T)$$

$$\therefore \boxed{A = A_0(1+\beta\Delta T)} \quad (\beta = 2\alpha)$$

β = Coefficient of areal expansion

$\beta = 2\alpha$

↳ Unit = $1/^\circ C$

↳ Dimension = $[K^{-1}]$

Change in area $= A - A_0$

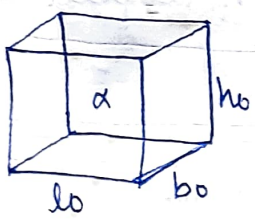
$= \beta A_0 \Delta T$

Fractional change in Area $= \frac{\Delta A}{A_0}$

$= \frac{\beta A_0 \Delta T}{A_0} = \beta\Delta T$

$= 2\alpha\Delta T$

Volumetric Expansion:



At $T^\circ C \Rightarrow V_0 = l_o b_o h_o$

At $(T_0+\Delta T)^\circ C \Rightarrow V = lbh$

$$V = l_o b_o h_o(1+\alpha\Delta T)^3$$

γ = Coefficient of volumetric expansion

$\gamma = 3\alpha$

$$V = V_0(1+\alpha\Delta T)^3$$

$$V = V_0(1+3\alpha\Delta T)$$

$$\boxed{V = V_0(1+\gamma\Delta T)} \quad (\gamma = 3\alpha)$$

Change in Volume $= V - V_0$

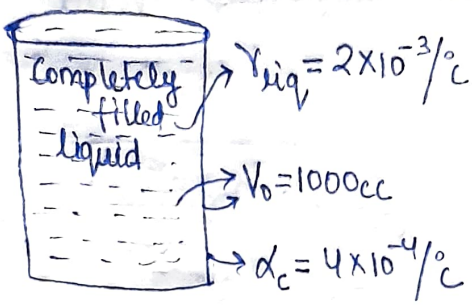
$= \gamma V_0 \Delta T$

Fractional change in volume $= \frac{\Delta V}{V_0} = \frac{\gamma V_0 \Delta T}{V_0} = \gamma\Delta T = 3\alpha\Delta T$

Note:

$$\boxed{\alpha : \beta : \gamma = 1 : 2 : 3}$$

Q

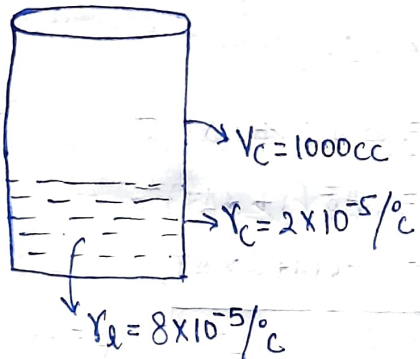


If T↑ by 20°C, find the volume of liquid which overflows.

$V_{overflow} = \Delta V_l - \Delta V_c$
$= V_0\gamma_l\Delta T - V_0\gamma_c\Delta T$
$= V_0(\gamma_l - \gamma_c)\Delta T$
$= V_0(\gamma_l - 3\alpha)\Delta T$

$V_{overflow} = 1000cc(2\times10^{-3} - 3\times4\times10^{-4})\times20$
$= 1000cc(0.8\times10^{-3})\times20 = 16cc$

Q



Find initial volume of liquid, if empty volume of container doesn't change with temperature.

Method 1:

$V_{empty} = V_C' - V_l'$
$= 1000(1+\gamma_C\Delta T) - V_l(1+\gamma_l\Delta T)$
$= (1000 - V_l) + (1000\gamma_C - V_l\gamma_l)\Delta T$

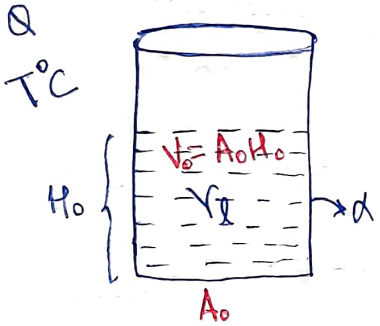
If volume doesn't change with temp.
$\Rightarrow 1000\gamma_C - V_l\gamma_l = 0 \Rightarrow 1000\gamma_C = V_l\gamma_l$

$V_l = \frac{1000\times2\times10^{-5}}{8\times10^{-5}} = 250cc$

Method 2:

$\Delta V_C = \Delta V_l$
$V_C\gamma_C\Delta T = V_l\gamma_l\Delta T$
$V_l = \frac{V_C\gamma_C}{\gamma_l} = 250cc$

Q

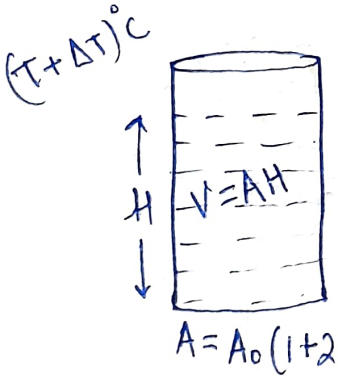


Find height of liquid column if temperature is increased by ΔT°C.

$V = V_0(1+\gamma\Delta T)$
$AH = A_0H_0(1+\gamma\Delta T)$
$A_0(1+2\alpha\Delta T)H = A_0H_0(1+\gamma\Delta T)$

$$\boxed{H = \frac{H_0(1+\gamma\Delta T)}{(1+2\alpha\Delta T)}}$$

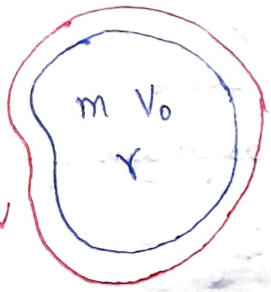
$T\uparrow \Rightarrow \Delta T = +ve$
$\gamma = 2\alpha \Rightarrow H = H_0$
$\gamma > 2\alpha \Rightarrow H > H_0$
$\gamma < 2\alpha \Rightarrow H < H_0$



OR $H = H_0(1+\gamma\Delta T)(1+2\alpha\Delta T)^{-1}$
$H = H_0(1+\gamma\Delta T)(1-2\alpha\Delta T)$

$$\boxed{H = H_0(1+(\gamma-2\alpha)\Delta T)}$$

Variation of density with temperature:



At $T°C \Rightarrow \rho_0 = \frac{m}{V_0}$

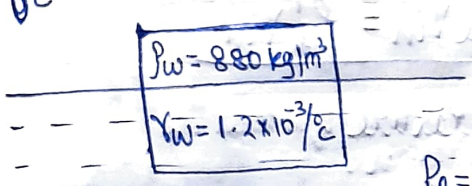
At $(T+\Delta T)°C \Rightarrow \rho = \frac{m}{V} = \frac{m}{V_0(1+\gamma\Delta T)}$

$$\boxed{\rho = \frac{\rho_0}{(1+\gamma\Delta T)}}$$ ← use this

Also, $\rho = \rho_0(1+\gamma\Delta T)^{-1}$

$$\boxed{\rho = \rho_0(1-\gamma\Delta T)}$$

Q



Find the temperature at which wood will ~~first~~ just sink in liquid.

To sink $\Rightarrow \rho_l' = \rho_w'$

$$\frac{\rho_l}{1+\gamma_l\Delta T} = \frac{\rho_w}{1+\gamma_w\Delta T}$$

$$= \rho_l + \rho_l\gamma_w\Delta T = \rho_w + \rho_w\gamma_l\Delta T$$

$$\Rightarrow \Delta T = \frac{\rho_l - \rho_w}{\rho_w\gamma_l - \rho_l\gamma_w}$$

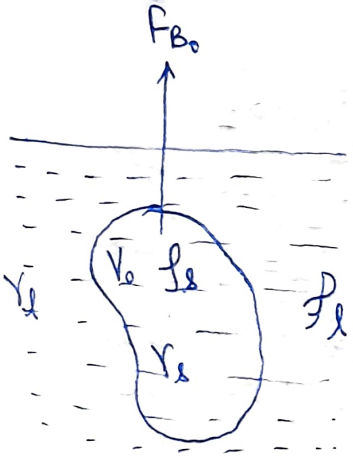
$$\Delta T = \frac{900-880}{880\times1.5\times10^{-3} - 900\times1.2\times10^{-3}}$$

$$\Delta T = \frac{20\times10^3}{880\times1.5 - 900\times1.2} = \frac{20\times10^3}{1320-1080}$$

$$\Delta T = \frac{20000}{240} = 83.33°C \quad \therefore \boxed{T_f = 83.33°C}$$

Variation of Buoyancy force with temperature:

Case①: Solid is completely immersed in liquid:



At $T°C \Rightarrow F_{B_0} = m_l g = \rho_l V_0 g$

At $(T+\Delta T)°C \Rightarrow F_B = \rho_l' V' g$

$$F_B = \frac{\rho_l V_0(1+\gamma_s\Delta T)g}{(1+\gamma_l\Delta T)} = \frac{F_{B_0}(1+\gamma_s\Delta T)}{(1+\gamma_l\Delta T)}$$

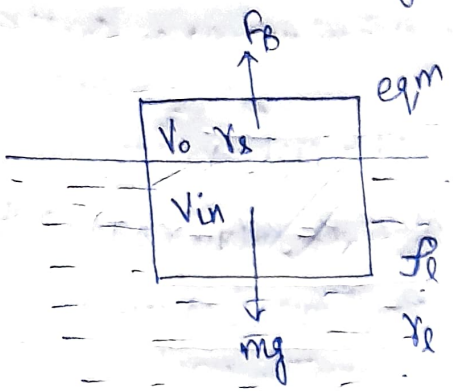
$$\boxed{F_B = \frac{F_{B_0}(1+\gamma_s\Delta T)}{(1+\gamma_l\Delta T)}}$$

If ΔT = +ve $\Rightarrow$ if $\gamma_s < \gamma_l \Rightarrow F_B < F_{B_0}$

if $\gamma_s = \gamma_l \Rightarrow F_B = F_{B_0}$

if $\gamma_s > \gamma_l \Rightarrow F_B > F_{B_0}$

Case ②: Solid is floating in liquid:



For solid to float ⇒

F_B = weight of solid

$F_B = mg$ ⇒ independent of temp.

Also, $F_B = m_l g = \rho_l V_{in} g$

$\Rightarrow \rho_l V_{in} g = mg \Rightarrow \rho_l V_{in} = m = \text{const}$

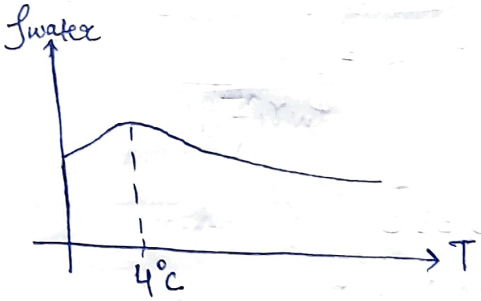
$\therefore \boxed{\rho_l V_{in} = \rho_l' V_{in}'}$

Variation of density of water with temperature:

$0°C \leq T < 4°C \Rightarrow \gamma_{water} = -ve \Rightarrow T\uparrow\ V\downarrow\ \rho\uparrow$

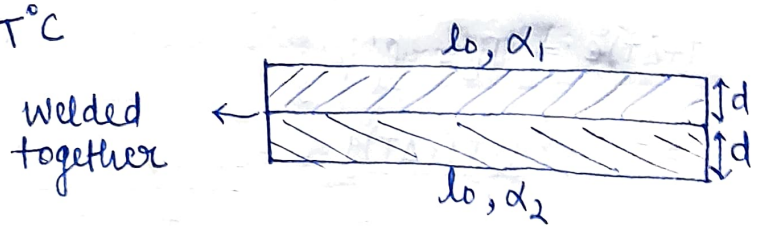
$T > 4°C \Rightarrow \gamma_{water} = +ve \Rightarrow T\uparrow\ V\uparrow\ \rho\downarrow$

* Water is most dense at 4°C.



Bimetallic strip:

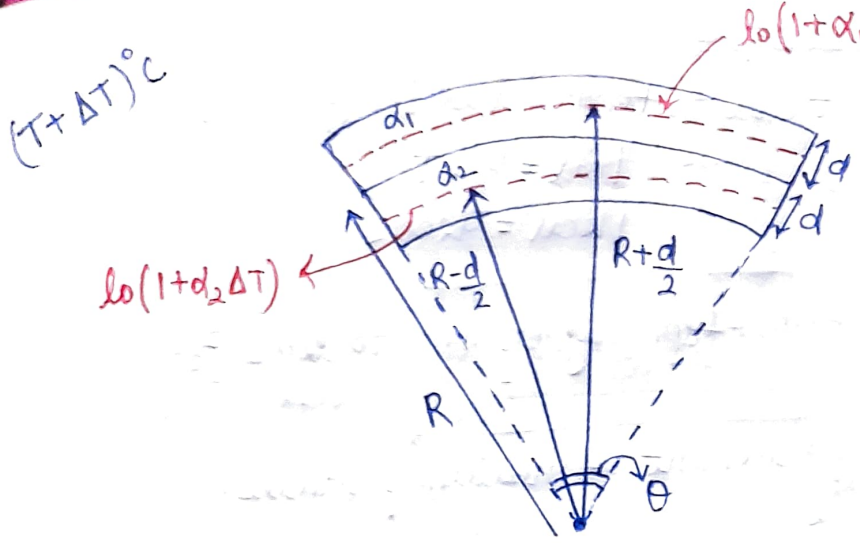
↳ 2 metals



$(\alpha_1 > \alpha_2)$

When we raise temperature, both strips tend to increase their length. As $\alpha_1 > \alpha_2$, so strip with α_1 will have larger increment than strip with α_2.

Also the strip are welded together, so finally they make a shape of circular arc.



$$\left(R+\frac{d}{2}\right)\theta = l_0(1+\alpha_1\Delta T) \quad \text{—①}$$

$$\left(R-\frac{d}{2}\right)\theta = l_0(1+\alpha_2\Delta T) \quad \text{—②}$$

Dividing ① and ②

$$\frac{R+\frac{d}{2}}{R-\frac{d}{2}} = \frac{1+\alpha_1\Delta T}{1+\alpha_2\Delta T}$$

$$= \frac{2R+d}{2R-d} = \frac{1+\alpha_1\Delta T}{1+\alpha_2\Delta T}$$

$$\Rightarrow 2R+d+2R\alpha_2\Delta T+d\alpha_2\Delta T = 2R-d+2R\alpha_1\Delta T-d\alpha_1\Delta T$$

$$= 2R\Delta T(\alpha_1-\alpha_2) = d(\alpha_1+\alpha_2)\Delta T + 2d$$

$$R = \frac{d(\alpha_1+\alpha_2)}{2(\alpha_1-\alpha_2)} + \frac{d}{(\alpha_1-\alpha_2)\Delta T}$$

$$\approx \boxed{R = \frac{d}{(\alpha_1-\alpha_2)\Delta T}}$$

Now, Subtracting ② from ①

$$\theta d = l_0(\alpha_1-\alpha_2)\Delta T$$

$$\boxed{\theta = \frac{l_0(\alpha_1-\alpha_2)\Delta T}{d}}$$